GANESH RAMESH

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EXECUTIVE SUMMARY

A seasoned Technical Director and Machine Learning Executive with two decades of experience successfully taking high-stakes AI products from concept to large-scale production. I specialize in defining the technical strategy, architecting, and leading teams to deliver cutting-edge solutions in Search, Generative AI and Conversational Agent systems.
My expertise spans the entire lifecycle of AI products and is strategically focused on the core pillars of modern AI products:

- **Agent Architecture:** Designing and shipping robust, context-aware Generative AI agents powered by LLMs, RAG, and sophisticated memory systems.
- **Search & Retrieval:** Architecting and deploying high-relevance, large-scale ML-Rankers for products like Maps and Enterprise Search.
- **Evaluation & Measurement:** Defining the analytics and metric framework necessary to quantify product effectiveness, accuracy, and business impact for new AI paradigms.
- **Strategic Leadership:** Proven ability to manage multi-year projects, chair internal technical summits, and actively mentor engineers across all levels.

CORE TECHNICAL COMPETENCIES

- **Generative AI & LLMs:** Agent Architecture, RAG Applications, Prompt Engineering, LLM Benchmarking & Evaluation Frameworks, Low-Cost Routing Strategies, Compact Model Fine-Tuning for Summarization.
- **Search & Information Retrieval:** Architecting and shipping large-scale Machine Learning Rankers (e.g., Maps Address Ranker), Enterprise Search Relevance, Query Understanding (Siri), and Information/Navigational Query Analysis.
- **Data Science & Evaluation:** Defining and leading product-level analytics and measurement for AI effectiveness, Designing performance metrics for stability and accuracy of forecasting, Data Audits, Prediction Modeling with significant lifts (e.g., 20% lift in clicker prediction).
- **Technical Leadership & Mentorship:** Lead for multiple complex product modules , Cross-functional project management for multi-year initiatives, Active mentor for engineers across all levels, and organizer/chair for internal ML summits.

EDUCATION

Postdoctoral Fellow, Database Systems Lab — **11/2003 – 10/2005**
University of British Columbia

Ph.D. Computer Science. — **10/2003**
University at Albany, State University of New York(SUNY)

M.S. (by Research) in Computer Science and Engineering: **06/1997**
Indian Institute of Technology, Madras (Now Chennai)

B.E. Computer Science and Engineering **05/1994**
University of Madras

PROFESSIONAL EXPERIENCE

Samsung Research America Mountain View, CA
(Consulting) Gen AI Engineer November 2024 - Present

- **Technical Direction for Health Agent and Leadership:** Provided technical direction for the core modules and analytics strategy of the Samsung Digital Health Assistant
- **Architected and Shipped the Conversation Memory Module:** Led the end-to-end development and delivery of the agent's memory system, storing long- and short-term conversation history and artifacts. This crucial module provides context for response grounding via RAG applications and advanced prompt engineering.
- **Evaluation and Analytics Strategy:** Leading the definition and implementation of the analytics and measurement framework to quantify the effectiveness and accuracy of the health assistant.
- **Supervised Research in Compact Models:** Mentored an intern to successfully develop fine-tuned small language models for conversation summarization that achieved comparable performance to larger models, resulting in a work currently under peer review.

Ema Unlimited Mountain View, CA
Founding ML Tech Lead December 2023 - August 2024

- **Defined Generative AI Foundations:** Served as the ML lead, responsible for architecting and developing the core generative AI technology that powered all downstream product applications.
- **Strategic LLM Evaluation and Cost Reduction:** Developed benchmarking frameworks for various LLM tasks across the product ecosystem. Engineered sophisticated routing strategies to achieve a low-cost, high-accuracy inference pipeline.
- **Agent Traceability and Response Strategy:** Developed robust design patterns for ensuring traceability and auditability of conversations. Explored and implemented smart composition strategies for LLM-based response generation.

Apple Inc. Cupertino, CA

Principal Research Engineer, AIML March 2022 - November 2023
Siri Query Understanding (AIML) - Tech Lead

- **Model Explainability & Debugging Tool:** Developed and shipped a gradient-based internal tool that provided crucial explainability and insight into model predictions. This tool was extensively used for production debugging and improving data quality.
- **High-Impact Model Deployment:** Successfully shipped fine-tuned address and app store parser models for multiple markets.

Principal Research Engineer, Maps Search — August 2017 - February 2022
Maps Search Architect/ML Lead for Address Ranker

- **Architected and Shipped the Production ML Ranker:** Led the successful deployment of the Machine Learning ranker for address search.
- **Strategic Project Management:** Directed the end-to-end, two-year project, including back-end pipeline architecture for offline model generation , outbound communication, and extensive cross-team feature development efforts.

Management and Leadership:

1. **Apple Mentor:** Technical mentor for junior team members and career guidance for 5 engineers across all functions within Apple.
2. **ML Thought Leadership:** Chaired the Knowledge Bases and Search Track for the Apple ML summit for three consecutive years, significantly raising the quality and rigor of accepted work.

SAP — Palo Alto, CA
Principal Data Scientist — May 2016 - July 2017

- **Enterprise Search & Agent Development:** Led initiatives to improve enterprise search relevance and contributed to the development of intelligent procurement agents.
- **Strategic Data Consulting:** Served as a consultant for key projects, focusing on transforming personnel data and ensuring Data Privacy for Machine Learning applications.
- **Data Exploration:** Directed initial data exploration efforts essential for building features for intelligent procurement solutions

Yahoo! Inc. (Yahoo! GEMINI) — Sunnyvale, CA
Principal Research Engineer — October 2015 - May 2016

- **Search Quality Optimization:** Shipped two major algorithmic improvements to sponsored search ad quality methods, leveraging data-driven analyses to bridge the gap between advertiser expectation and system performance.
- **Metrics Design:** Developed data-driven metrics to engage with advertisers and promote the value of the Yahoo sponsored search serving system.

Technologies/Languages: Java, Python, Pig, Hive, Hadoop

Edmodo Inc. — San Mateo, CA
Principal Scientist/Engineer/Lead — November 2013 - August 2015

- **Content Ecosystem Leadership:** Led a team that successfully shipped a content recommendation service and an educational content ecosystem complete with ratings, reviews, and search/discovery capabilities.
- **Content Curation and Information Extraction:** Designed and built a set of educational crawling and information extraction tools that was used to crawl millions of educational content items for search and recommendations.
- **Matching Algorithm Innovation:** Designed and built a set of novel matching algorithms for educational objects, enabling fast, approximate matches that powered various recommendation services.

LinkedIn Corporation (Product Data Sciences) Mountain View, CA
Senior Research/Data Scientist April 2012 - November 2013

- **Invented Core Data Products:** Engineered and shipped two location-based products: the Business Traveler User Segment and the Inferred Location Update (part of the profile completion product).
- **High-Value Predictive Modeling:** Engineered features from session data to predict clickers, resulting in a 20% lift in prediction accuracy when integrating session-based features.
- **Patent Portfolio Contribution:** Filed 4 patents in the areas of Inferred Identity, Recommendation Systems, and Social Reputation for professional social networks.

Yahoo! Labs (Advertising Sciences) Santa Clara, CA
Senior Research Engineer August 2009 - March 2012

- **Large-Scale Forecasting System:** Engineered components of a state-of-the-art, large-scale supply forecasting system for complex multi-dimensional impression forecasting for Yahoo's Display Advertising platform.
- **Evaluation & Auditing Framework:** Designed a comprehensive set of performance metrics and engineered the data pipelines to measure stability and accuracy of guaranteed display advertising contracts. Conducted impactful data audits that validated serving algorithms and drove system improvements.

Microsoft Corporation(Windows Live Safety Platform), Antispam Redmond, WA
Software Development Engineer May 2007 - July 2009

- **High-Impact Spam Reduction System:** Researched and engineered a scalable system utilizing router reputation mined from Traceroute Data to filter spam. This deployment resulted in a measurable reduction of spam by 2% in a conservative production setting.
- **Data Collection Velocity Improvement:** Engineered a multi-threaded, generic network data collection tool (DNS, RDNS, Traceroute, Port scanners) that significantly accelerated data throughput by up to two orders of magnitude compared to baseline single-threaded methods.
- **Feature Exploration:** Investigated the impact of generic Natural Language Features extracted from graded mail to enhance spam classification model

Microsoft Corporation(Live Search) Redmond, WA
Software Development Engineer Jul 2006 - May 2007

- **Search Quality Assurance:** Shipped language-insensitive methods based on click logs to mine and monitor "Definitive Answers" lists, identifying potentially erroneous entries. These techniques substantially improved quality, impacting up to 10% of entries in many international markets.
- **Query Intent Analysis:** Conducted foundational investigation of query and click logs to distinguish between informational queries (seeking broad data) versus navigational queries (satisfied by a single URL).

Amazon.com, Customer Behaviour Group Seattle, WA.
Analyst Oct 2005 - Jun 2006

- **Core Machine Learning Model Deployment:** Designed, implemented, and shipped a scalable machine learning model to optimize advertisement content delivery. This system determined the best advertisements for display on Amazon associate websites.
- **Optimization and Tuning:** Provided support and iterative tuning for the model, focusing on improving the quality of recommendations and incorporating advanced phrasal features.

Database Systems Lab, University of British Columbia Vancouver, B.C.
Postdoctoral Research Fellow Nov 2003 - Oct 2005

- Research on "Satisfiability of Tree Pattern Queries" resulting in a publication in VLDB 2004.
- Research on "Data Privacy and Disclosure Risk Analysis" resulting in two conference papers in SIGMOD 2005 and ICDE 2007, a workshop paper in Secure Data Management 2005 and a journal paper in ACM-TKDD.

Avaya Labs Research Basking Ridge, New Jersey
Research Intern Aug 2001 - Nov 2001

- Research on multimedia indexing and clustering of news articles leading to development of hybrid clustering algorithms. This research led to three conference publications.

University at Albany, State University of New York Albany, New York
Research Assistant Sep 2000 - May 2003

- Research on Indexing and Data Access Methods for Data Mining and Distribution-Based Synthetic Database Generation. Led to one workshop publication (DMKD 2002) and one conference publication in PODS 2003.

Hewlett-Packard labs Research Palo Alto, California
Research Intern May 1999 - Aug 1999

- Research leading to the development of an association rule mining workbench to benchmark the performance of algorithms.

General Electric Corporate Research and Development Labs Schenectady, New York
Research Intern May 1998 - Aug 1998

- **Customer Retention Models:** Applied decision tree and logistic regression methods to develop customer retention models for GE Capital products. Successfully identified potential customers for further action.

TEACHING EXPERIENCE

Instructor:

- CPSC304 - Introduction to Relational Databases, University of British Columbia, Term: Summer 2004
- CSI402 - Systems Programming, University at Albany, SUNY, Terms: Summer 2000, 2003, 2003
- CSI210 - Introduction to Discrete Mathematics, University at Albany, SUNY, Term: Summer 2000

Teaching Assistant:

- CSI402 - *Algorithms and Data Structures*, Terms: Fall 1997, Fall 1999.
- CSI409 - *Theory of Computation*, Terms: Spring 1998, Spring 1999.
- CSI311 - *Principles of Programming Languages*, Terms: Spring 2002, Spring 2003.
- CSI333 - *Programming the Hardware/Software Interface*, Term: Fall 1998.
- CSI508 - *Database Systems*, Term: Spring 2000.

PROFESSIONAL SERVICE

- Member of the Faculty Search Committee for the Department of Computer Science at University at Albany (SUNY) from 1999 until 2001.
- Organizing committee member for the Simulation Day (October 2002), the Discrete Mathematics and Computer Science Day (DMCSD) (September 2002) and the Network Design Day (October 1998).
- **Program Committee Member** for several conferences in Data Mining and Management and for the ICDM conference since 2011.

PUBLICATIONS - Refereed Conferences and Workshops

1. Sean Chester, Bruce Kapron, Ganesh Ramesh, Gautam Srivastava, Alex Thomo and S. Venkatesh, **K-Anonymization of Social Networks by Vertex Addition**, in Proceedings II of 15^{th} East-European Conference on Advances in Databases and Information Systems (ADBIS), ADBIS 2011 Research Communications, 2011.
2. Shaofeng Bu, Laks V.S. Lakshmanan, Raymond Ng and Ganesh Ramesh, **Preservation of Patterns and Input-Output Privacy**. In IEEE International Conference on Data Engineering **ICDE**, Istanbul, Turkey, 2007.
3. Ganesh Ramesh, **Can Attackers Learn from Samples?** In 2^{nd} **VLDB** workshop on **Secure Data Management**, Trondheim, Norway, September 2005.
4. Ganesh Ramesh, Mohammed J. Zaki and William A. Maniatty, **Distribution-Based Synthetic Database Generation Techniques for Itemset Mining**. In 9^{th} International Database Engineering and Applications Symposium **IDEAS**, Montreal, Canada, July 2005.
5. Laks V.S. Lakshmanan, Raymond Ng and Ganesh Ramesh, **To Do or Not To Do; The Dilemma of Disclosing Anonymized Data**. In **ACM SIGMOD** Conference, Baltimore, Maryland, June 2005.
6. Laks V.S. Lakshmanan, Ganesh Ramesh, Wendy (Hui) Hwang and Jessica (Zheng) Zhao, **On Testing Satisfiability of Tree Pattern Queries**. In **VLDB** Conference, Toronto, Canada, August 2004.
7. Ganesh Ramesh, William A. Maniatty and Mohammed J. Zaki, **Feasible Itemset Distributions in Data Mining: Theory and Application**. In **ACM PODS** Conference, San Diego, California, June 2003.
8. Ganesh Ramesh, William A. Maniatty and Mohammed J. Zaki, **Indexing and Data Access Methods for Database Mining**. In ACM-SIGMOD Workshop on Data Mining and Knowledge Discovery **(DMKD)**, Madison, Wisconsin, June 2002.

9. Amit Bagga, Jianying Hu, Jialin Zhong and Ganesh Ramesh, **Multi-source Combined-Media Video Tracking for Summarization**. In 16th International Conference on Pattern Recognition **(ICPR)**, Quebec City, Canada, August 2002.

10. Ganesh Ramesh and Amit Bagga, **A Text-based Method for Detection and Filtering of Commercial Segments in Broadcast News**. In International Conference on Language Resources and Evaluation **(LREC)**, Las Palmas, Canary Islands, Spain, May 2002.

11. Amit Bagga, Breck Baldwin and Ganesh Ramesh, **A Methodology for Cross-Document Coreference Over Degraded Data Sources**. In Recent Advances in Natural Language Processing **(RANLP)**, Tzigov Chark, Bulgaria, September 2001.

PUBLICATIONS - Refereed Journals:

1. Sean Chester, Bruce M. Kapron, Ganesh Ramesh, Gautam Srivastava, Alex Thomo, S. Venkatesh, **Why Waldo befriended the dummy? k-Anonymization of social networks with pseudo-nodes**. In Social Network Analysis and Mining, Volume 3, Number 3, pages 381-399, 2013.

2. Laks V.S. Lakshmanan, Raymond Ng and Ganesh Ramesh, **On Disclosure Risk Analysis of Anonymized Itemsets in the Presence of Prior Knowledge**. In ACM Transactions on Knowledge Discovery in Data (TKDD), Volume 2, Number 3, 2008.

PATENTS

1. **U.S. Patent No. 8812690 - August 19, 2014:** Method and system to provide reputation scores for a social network member. Inventors: **Ganesh Ramesh**, Ramakrishna Vemuri

2. **U.S. Patent No. 9635116 - April 25, 2017:** Techniques for inferring a location. Inventors: Sathyanarayan Anand, **Ganesh Ramesh**, Alexis Blevins Baird

3. **U.S. Patent No. 9661039 - May 23, 2017:** Recommending resources to members of a social network. Inventors: Heyning Cheng, Navneet Kapur, **Ganesh Ramesh**

Employment Eligibility: United States Citizen.

REFERENCES: Available upon request.